

Recurrent Neural Networks

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A lot of the ideas in this lecture come from Andrej Karpathy's blog post on the Unreasonable Effectiveness of RNNs (<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>). Many of the images and animations were made by Adam Prügel-Bennett.

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Recurrent Neural Networks - Motivation

x : Jon and Antonia gave deep learning lectures

y : 1 0 1 0 0 0 0

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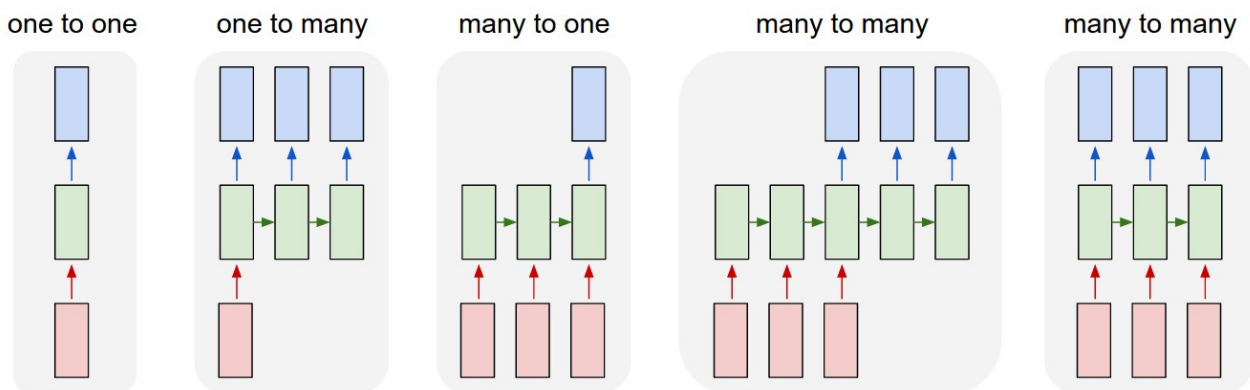
Recurrent Neural Networks - Motivation

x : $x^{(1)}$... $x^{(t)}$... $x^{(T_x)}$
 x : Jon ... Antonia ... lectures

y : $y^{(1)}$... $y^{(t)}$... $y^{(T_y)}$
 y : 1 ... 1 ... 0

In this example, $T_x = T_y = 7$ but T_x and T_y can be different.

Recurrent Neural Networks



Why Not a Standard Feed Forward Network?

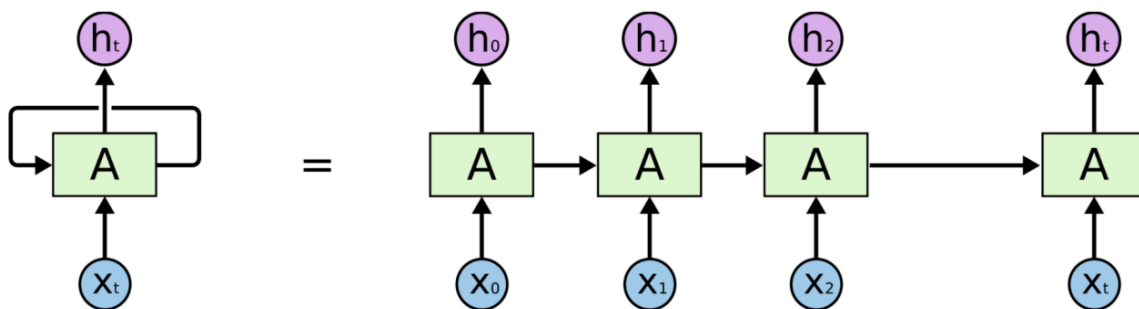
- For a task such as “Named Entity Recognition” a MLP would have several disadvantages
 - The inputs and outputs may have varying lengths

Image from <http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

- The features wouldn't be shared across different temporal positions in the network
 - * Note that 1-D convolutions can be (and are) used to address this, in addition to RNNs - more on this in a later lecture
- To interpret a sentence, or to predict tomorrow's weather it is necessary to remember what happened in the past
- To facilitate this we would like to add a feedback loop delayed in time

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Recurrent Neural Networks



- RNNs are a family of ANNs for processing sequential data
- RNNs have directed cycles in their computational graphs

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Recurrent Neural Networks

RNNs combine two properties which make them very powerful.

1. Distributed hidden state that allows them to store a lot of information about the past efficiently. This is because several different units can be active at once, allowing them to remember several things at once.
2. Non-linear dynamics that allows them to update their hidden state in complicated ways¹.

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Recurrent Neural Networks

RNNs are Turing complete in the sense they can simulate arbitrary programs².

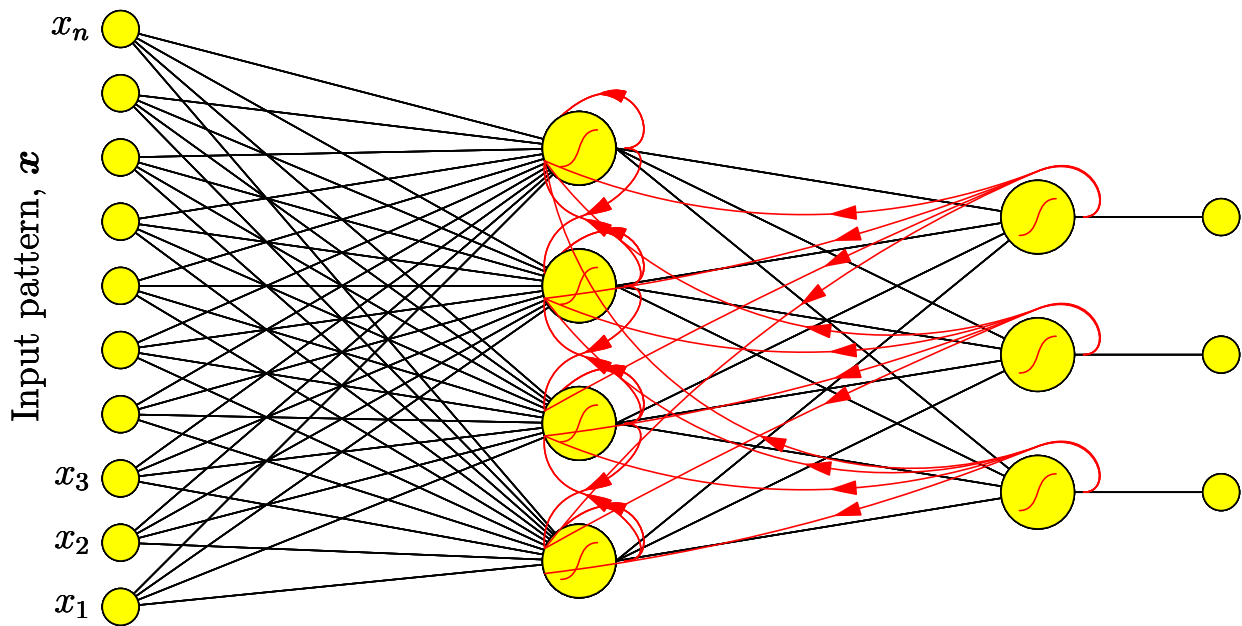
If training vanilla neural nets is optimisation over functions, training recurrent nets is optimisation over programs.

Image taken from <https://towardsdatascience.com>

¹Often said to be difficult to train, but this is not necessarily true - dropout can help with overfitting for example

²Don't read too much into this - like universal approximation theory, just because they can doesn't mean it's necessarily learnable!

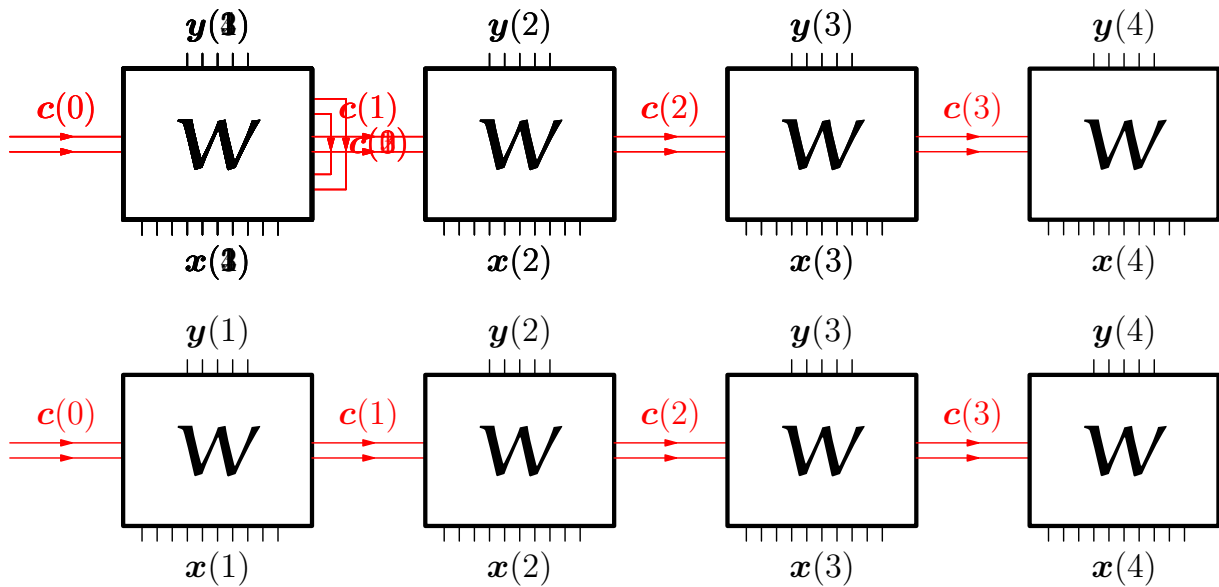
Recurrent Network



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Training Recurrent Networks

- Given a set of inputs $\mathcal{D} = ((x(t), y(t)) | t = 1, 2, \dots, T)$



- Minimise an error (here MSE, but your choice):

$$E(W) = \sum_{t=1}^T \|y(t) - f(x(t), c(t-1)|W)\|^2$$

- This is known as *back-propagation through time*

An RNN is just a recursive function invocation

- $y(t) = f(x(t), c(t-1)|W_f)$

- and the state $\mathbf{c}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{c}(t-1) | \mathbf{W}_g)$
- If the output $\mathbf{y}(t)$ depends on the input $\mathbf{x}(t-2)$, then prediction will be

$$\mathbf{f}(\mathbf{x}(t), \mathbf{g}(\mathbf{x}(t-1), \mathbf{g}(\mathbf{x}(t-2), \mathbf{g}(\mathbf{x}(t-3) | \mathbf{W}_g) | \mathbf{W}_g) | \mathbf{W}_g) | \mathbf{W}_f)$$

- it should be clear that the gradients of this with respect to the weights can be found with the chain rule

What is the state update $\mathbf{g}()$?

- It depends on the variant of the RNN!
 - Elman
 - Jordan
 - LSTM
 - GRU

Elman Networks (“Vanilla RNNs”)

$$\begin{aligned}\mathbf{h}_t &= \sigma_h(\mathbf{W}_{ih}\mathbf{x}_t + \mathbf{b}_{ih} + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_{hh}) \\ \mathbf{y}_t &= \sigma_y(\mathbf{W}_y\mathbf{h}_t + \mathbf{b}_y)\end{aligned}$$

- σ_h is usually tanh
- σ_y is usually identity (linear) – the $\mathbf{y}$ ’s could be regressed values or logits
- the state $\mathbf{h}_t$ is referred to as the “hidden state”
- the output at time t is a projection of the hidden state at that time
- the hidden state at time t is a summation of a projection of the input and a projection of the previous hidden state

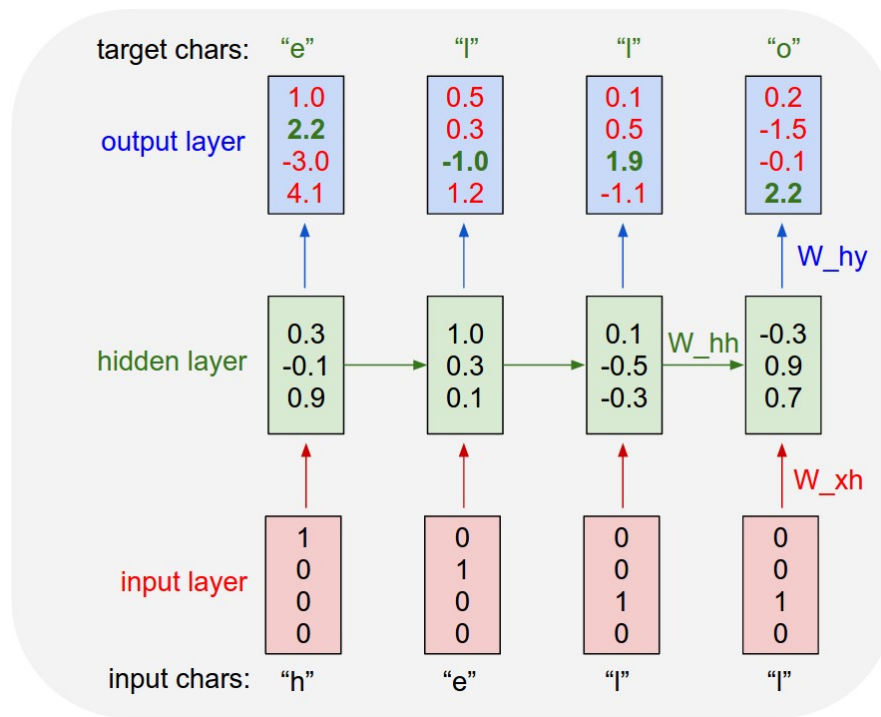
Going deep: Stacking RNNs

- RNNs can be trivially stacked into deeper networks
- It’s just function composition: [1em] $\mathbf{y}(t) = \mathbf{f}_2(\mathbf{f}_1(\mathbf{x}(t), \mathbf{c}_2(t-1) | \mathbf{W}_1), \mathbf{c}_2(t-1) | \mathbf{W}_2)$ [1em]
- The output of the inner RNN at time t is fed into the input of the outer RNN which produces the prediction $\mathbf{y}$
- Also note: RNNs are most often not used in isolation - it’s quite common to process the inputs and outputs with MLPs (or even convolutions)

Example: Character-level language modelling

- We’ll end with an example: an RNN that learns to ‘generate’ English text by learning to predict the next character in a sequence
- This is “Character-level Language Modelling”

Example: Character-level language modelling



Training a Char-RNN

- The training data is just text data (e.g. sequences of characters)
- The task is unsupervised (or rather self-supervised): given the previous characters predict the next one
 - All you need to do is train on a reasonable sized corpus of text
 - Overfitting could be a problem: dropout is very useful here

Sampling the Language Model

- Once the model is trained what can you do with it?
- if you feed it an initial character it will output the logits of the next character
- you can use the logits to select the next character and feed that in as the input character for the next timestep
- how do you 'sample' a character from the logits?
 - you could pick the most likely (maximum-likelihood solution), but this might lead to generated text with very low variance (it might be boring and repetitive)
 - you could treat the softmax probabilities defined by the logits as a categorical distribution and sample from them
 - * you might increase the 'temperature', T , of the softmax to make the distribution more diverse (less 'peaky'): $q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$

A lot of the ideas in this lecture on the input $c(t-1)$, $g(x_i(x(t-2), g(t-1) - W))$ Iged snllhomitpon" are Mnt Net) th

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Image from <http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

- Sampled from a single layer RNN³.

³LSTM, 128 dim hidden size, with linear input projection to 8-dimensions and output to the number of characters (84). Trained on the text of these slides for 50 epochs.